MIL-H-37435A (DSA-DM)
20 July 1976
SUPERSEDING
MIL-H-37435 (DSA-DM)
13 January 1976

MILITARY SPECIFICATION

HEMORRHOIDAL SUPPOSITORIES WITH HYDROCORTISONE ACETATE

This specification is approved for use by the Defense Personnel Support Center, Defense Supply Agency, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers Hemorrhoidal Suppositories with Hydrocortisone Acetate (See 6.2).

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 6505

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SPECIFICATIONS

FEDERAL

PPP-C-186 - Containers, Packaging and Packing for Drugs, Chemicals and Pharmaceuticals. (See 6.3)

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procureing activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

U. S. DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE, FOOD AND DRUG ADMINISTRATION

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder.

(Application for copies should be addressed to the Food and Drug Administration, Washington, DC 20204.)

U. S. PHARMACOPEIAL CONVENTION, INC.

The United States Pharmacopeia.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

National Formulary.

(Application for copies should be addressed to the Mack Publishing Company, Easton, PA 18042.)

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3. REQUIREMENTS

3.1 Material. Shall be a rectal suppository, with each suppository containing the following;

Hydrocortisone Acetate, USP	10 mg
Belladonna Extract (0.19 mg equivalent total Alkaloids)	15 mg
Ephedrine Sulfate	3 mg
Zinc Oxide	176 mg
Boric Acid	543 mg
Bismuth Oxy-iodide	30 mg
Bismuth Subcarbonate	146 mg
Peruvian Balsam	30 mg
Cocoa butter and Yellow wax	qs

3.1.1 Batch formulation sheet. The batch formulation sheet (raw material ingredient weighing specifications) for each lot shall be reviewed to determine conformance with the formulation specified in 3.1 of this document.

3.2 Finished product.

3.2.1 Assay for hydrocortisone acetate. The suppository shall assay to contain not less than 95.0 percent and not more than 110.0 percent of the required amount of hydrocortisone acetate when determined as specified in 4.7.1.

3.2.2 Uniformity of mixture. The suppository shall show little or no unmixed color or uncolored areas when cut longitudinally. The suppositories may be slightly mottled or speckled.

3.2.3 Softening time. Softening time for the suppository shall be not more than 15 minutes when determined as specified in 4.7.2.

3.2.4 Weight variation. Individually weigh 20 suppositories, and determine the average weight. No individual suppository shall deviate from the calculated average weight by more than ± 10 percent.

3.2.5 Fat soluble extract. The fat soluble extract shall be 69.0 percent ± 10 percent when determined as specified in 4.7.3.

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3.2.6 Accelerated aging test. The suppository shall show no rainbow effect, streaking, color separation, or crystallization on the surface, when subjected to test specified in 4.7.4.

3.3 Ingredients.

3.3.1 USP ingredients. The Hydrocortisone Acetate, Ephedrine Sulfate, Zinc Oxide and Cocoa Butter used in the manufacture of the suppositories shall comply with the tests, standards, and requirements of the USP, including any supplements or revisions thereto.

3.3.2 NF ingredients. The Belladonna Extract, Boric Acid and Yellow Wax used in the manufacture of the suppositories shall comply with the tests, standards, and requirements of the NF, including any supplements or revisions thereto.

3.3.3 Bismuth Oxy-iodide. The Bismuth Oxy-iodide (oxyidodide base) shall comply with the following requirements.

3.3.3.1 Identity. The bismuth, iodide, and carbonate shall be identified using the "General identification Tests" of the USP.

3.3.3.2 Assay for iodine. The bismuth oxy-iodide shall contain not less than 5.7 percent and not more than 6.6 percent of iodine when assayed as specified in 4.7.5.

3.4 Other ingredients. The Bismuth Subcarbonate and Peruvian Balsam shall be of a grade suitable for use in the suppositories.

3.5 Delivery. Not more than 4 months shall have elapsed from the date of manufacture of the product to the date of delivery to the Government.

3.6 Workmanship. The material and its containers shall be free from defects which detract from their appearance or may impair their serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and, (i) as to any expiration dated item, for the expiration dating period specified by the contractor for such item, or such longer period as may be required by regulation of any federal agency; or (ii) as to non-dated items, for not less than 3 years after delivery of the item to the Government.

4.1.2 No company supplying any ingredient(s) to the contractor will be considered an acceptable facility for the performance of any inspection requirements specified herein.

4.2 Lot. For purposes of this specification, a lot, batch, or control is that single, uniform, and homogeneous quantity of suppositories produced from one formulation, subjected to the same compounding and manufacturing operation, and filled into final containers.

4.3 Sampling. Sampling shall be conducted in accordance with MIL-STD-105.

TABLE I. Sampling.

For visual examination	Inspection level	AQL (percent defective)	Unit of product
Major	S-2	1.0	Filled immediate container
Minor	II	2.5	Filled immediate container

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4.3.1 For raw material and end item testing. Noncompliance with any of the following is cause for rejection.

TABLE II. Criteria for raw material and end item testing.

Characteristic	Requirement paragraph	Inspection level 1/	Minimum sample
Assay for hydrocortisone acetate	3.2.1	S-2	5
Uniformity of mixture	3.2.2	S-2	5
Softening time	3.2.3	S-2	5
Weight variation	3.2.4	S-2	20
Fat soluble extract	3.2.5	S-2	5
Accelerated aging test	3.2.6	S-2	12
Ingredients 2/			
Bismuth oxy-iodide	3.3.3		

1/ Where possible, the sample shall be used for two or more tests.
2/ Sufficient quantities of each of raw materials shall be selected from each lot to perform tests indicated in applicable test methods. Failure of any sample to comply with test requirements shall be cause for rejection of the lot represented.

4.4 Examination. The finished suppositories shall be examined to determine compliance with all requirements of this specification. Nonconformance will be permitted to the extent indicated in 4.3 and 4.3.1.

4.4.1 Classification of defects. Examination will be conducted in accordance with the following classification of defects:

TABLE III. Classification of defects.

Categories	Defects 1/
Major	
101	Suppository not free from foreign matter.
102	Suppository not free of crystal formation.
103	Suppository not homogeneous.
104	Suppository not free of breaks, pits or other irregularities in appearance.

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TABLE III. Classification of defects. (continued)

Categories	Defects 1/
Minor	
201	Exterior of immediate container is not free from contents or foreign matter.
202	Label not free from defacing marks and tears.

1/ Inspection is not restricted to the classified possible defects listed above.

4.5 Packaging inspection. The inspection of the intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.6 Labeling inspection. Sampling shall be in accordance with MIL-STD-105 and the unit of product shall be the unit of issue. The inspection lot shall be the number of units of issue of each batch, lot, control number, etc., offered for inspection at one time. The inspection level for labeling shall be S-2 and the acceptance number shall be zero.

4.7 Tests.

4.7.1 Assay for hydrocortisone acetate.

Reagents and Standards.

Ethyl alcohol, absolute.
n-Hexane
Sulfuric acid solution, 62 percent v/v. Carefully dilute 310 ml of reagent grade concentrated sulfuric acid to 500 ml with distilled water.

Acid-phenylhydrazine reagent. Dissolve 65 mg of phenylhydrazine hydrochloride in 100 ml of the 62 percent sulfuric acid solution.

Hydrocortisone standard solutions.

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Stock solution. Dissolve 20 mg of hydrocortisone acetate USP reference standard in 100 ml of absolute ethyl alcohol.

Working solution. Dilute 5 ml of the stock solution to 50 ml with absolute ethyl alcohol. Each ml contains 0.020 mg of hydrocortisone acetate.

Procedure

Add 100 ml of absolute ethyl alcohol to 5 weighed suppositories contained in a 250 ml volumetric flask and warm on a hot water bath until the suppositories are melted and dispersion is complete. Cool the flask to room temperature and dilute to volume with absolute ethyl alcohol. Stopper and mix thoroughly. Filter the mixture through Whatman No. 1 filter paper, discarding the first 25 ml of filtrate. Dilute 10 ml of a succeeding portion of the clear filtrate to 50 ml with absolute ethyl alcohol. Transfer 7 ml of this dilution to a 50 ml separatory funnel, add 3 ml of distilled water and mix. Wash the mixture twice with 5 ml portions of n-hexane, discarding the hexane washes. Collect the alcohol-aqueous solution in a 50 ml beaker and carefully evaporate to dryness employing mild heat and an air-stream. Dissolve the residue in 14 ml of absolute ethyl alcohol to yield a theoretical hydrocortisone acetate concentration of 0.020 mg/ml. Transfer, in triplicate, 1 ml aliquots of both sample and standard solutions to separate 10 ml test tubes. Add 3 ml of acid-phenylhydrazine reagent to two of the samples and two of the standard tubes and 31 of the 62 percent sulfuric acid solution to each of the two remaining tubes (sample and standard blank). Immerse all tubes for 20 minutes in a water bath maintained at 60°C. Cool all tubes under running tap water to room temperature and determine the absorbance of both solutions in 1 cm cells at 410 nm with a suitable spectrophotometer. As a reference liquid to zero the spectrophotometer, use a solution of 1 ml of absolute ethyl alcohol and 3 ml of the acid-phenylhydrazine reagent carried through the heating and cooling treatment.

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Calculation:

$$\frac{Au}{As} \times \frac{Cs}{I} \times \frac{2500}{I} \times \frac{S}{Wu} = X$$

$$\frac{X}{10} \times \frac{100}{I} = \text{percentage of hydrocortisone acetate}$$

Where:

Au = absorbance of sample at 410 nm
As = absorbance of standard at 410 nm
Cs = concentration of working standard in mg/ml
S = average suppository weight in mg
Wu = sample weight (5 suppositories) in mg
X = mg of hydrocortisone acetate per suppository

4.7.2 Softening time.

Place a suppository (which has been kept refrigerated for not less than two hours immediately prior to test) blunt end up, into a test tube of suitable size. Impose a flat weight 1/, which is connected to a rod 2/ at its top center, directly on the suppository. Position the test tube in a 38°C water bath and determine the time required for the weight to fall 20 mm.

Not less than five (5) suppositories shall be used for the above test.

4.7.3 Fat soluble extract.

Weigh accurately one suppository and transfer it to an extraction thimble. Extract completely with chloroform (about 6 hours). Transfer the chloroform extract to a tared dish and evaporate to dryness. Dry the residue to constant weight at 100°C., cool in a desiccator and weigh.

Calculation:

$$\frac{\text{Weight residue}}{\text{Weight sample}} \times 100 = \text{percentage of fat soluble extract}$$

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4.7.4 Accelerated Aging Test.

Place twelve (12) foil wrapped suppositories horizontally in a Petri dish, cover with a Petri dish cover and place in an oven at $43^{\circ} \pm 2^{\circ}C$. At the end of one (1) hour, remove the Petri dish and allow the suppositories to cool at room temperature. When cool, place the dish into a refrigerator to solidify the suppositories, unwrap, and examine.

NOTE: The formation of cavities or white spots in the suppositories subjected to the above test is not cause for rejection.

4.7.5 Assay for iodine in bismuth oxy-iodide.

Weigh accurately about 5 Grams of sample (powder) and transfer to a 250 ml Erlenmeyer flask. Add 30 ml distilled water and agitate thoroughly and add 10 ml thoroughly wetted. Add 50 ml 0.1N $AgNO_3$, agitate thoroughly and add 10 ml HNO_3. Heat the mixture on a steam bath and agitate until the reaction is complete (precipitate is yellow in color). Cool and titrate the excess $AgNO_2$ with 0.1N thiocyanate, using 2 ml of ferric ammonium sulfate test solution as the indicator, to the first appearance of a red-brown color. Each ml 0.1N $AgNO_3$ is equivalent to .01269 Gram iodine. Calculate percent iodine.

1/ Weight and rod = 51 Gram
2/ Rod calibrated in mm.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with all applicable requirements of PPP-C-186 (See 6.3) and as specified herein.

5.2 Packaging. Packaging shall be level A or C, as specified (See 6.1).

5.2.1 Unit of issue. One box containing 12 suppositories, as specified, constitutes one unit of issue.

5.2.2 Level A.

5.2.2.1 Immediate container. One suppository shall be in a commercially available immediate container considered acceptable by the Food and Drug Administration (FDA) for the product contained therein which shall be of such compatibility that neither the contents nor the container are altered in any way by each other.

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5.2.2.2 Unit packaging. Each unit shall be packaged as specified in 5.2.5 of PPP-C-186.

5.2.2.3 Intermediate packaging. Twelve (12) units of issue shall be packaged as specified in 5.2.7.1 of PPP-C-186.

5.2.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.3 Packing. Packing shall be level A, B or C, as specified (see 6.1).

5.3.1 Level A. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.4 of PPP-C-186.

5.3.2 Level B. One hundred forty-four (144) units of issue shall be packed as specified in 5.4.3.1.1 of PPP-C-186.

5.3.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.4 Marking.

5.4.1 Labeling. Labeling shall be in accordance with the requirements of the Federal Food, Drug and Cosmetic Act and shall include the information required below:

5.4.1.1 Immediate container. Each immediate container label shall bear the following information:

(a) label information in accordance with commercial practice.

5.4.1.2 Unit package. Each unit package label shall bear the following information:

(a) label information in accordance with commercial practice.

(b) the National Stock Number.

(c) the date of manufacture prefixed by "MFD".

NOTES: (1) The National Stock Number shall not be preceded by any prefix.

(2) Strip labeling may be used.

(3) Label adhesion test shall not apply.

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5.4.2 Intermediate package. Each intermediate package shall be marked as specified in paragraph 5.5.3 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"STORE BETWEEN 2° - 8°C (35° to 46°F.)"

5.4.3 Exterior container. Exterior containers shall be marked as specified in 5.5.4 of PPP-C-186. Date of manufacture shall be shown in lieu of date packed. Marking shall include the legend:

"STORE BETWEEN 2° - 8°C (35° to 46°F.)"

NOTE: The unrefrigerated shipping time for this item shall be not more than 288 hours (12 days).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.

(b) National Stock Number.

(c) Selection of applicable levels of packaging and packing (See 5.2 and 5.3).

6.2 This specification covers the following item:

National Stock Number	Item Identification
6505-00-723-5015	HEMORRHOIDAL SUPPOSITORIES WITH HYDROCORTISONE ACETATE, 12s

6.3 All references to PPP-C-186 refer to Interim Federal Specification PPP-C-00186B which is mandatory for use until superseded by a revision.

Preparing activity:
Defense Supply Agency - DM

Project No. 6505-1762

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DOCUMENT IDENTIFIER AND TITLE	
MIL-H-37435A (DSA-DM)	HEMORRHOIDAL SUPPOSITORIES WITH HYDROCORTISONE ACETATE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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